

Dataset link:

<https://www.kaggle.com/datasets/anlgrbz/student-demographics-online-education-dataoulad/data>

Principles: *Transparency & Explainability* => Çağrı Altınüzengi

Fairness & Non-Discrimination => Agit Sefkan Balcı

Awareness & Literacy => Yasemin Gunindi

Privacy Concerns => Edoardo Balzano

Introduction

This report focuses on an education-domain AI use case: predicting student outcomes (e.g., pass, fail, or withdrawal) and triggering “early warning” interventions using learning analytics data, specifically the Open University Learning Analytics Dataset (Kuzilek et. al., 2017). The goal is to identify potential benefits, harms, bias risks, privacy issues, and oversight gaps that may arise when institutions use predictive models to classify students as “at risk” and act on those predictions. We selected four ethical principles that map naturally onto the main ethical pressure points of learning analytics systems: Transparency & Explainability, Fairness & Non-Discrimination, Awareness & Literacy and Privacy Concerns. These principles align with widely used trustworthy AI frameworks that emphasize transparency, fairness, and human oversight, particularly in high-stakes domains where automated decisions can affect individuals’ life opportunities, such as education.

One of the most important factors that contribute to this issue of ethics is the complexity and opacity of machine learning algorithms (Gunasekara et. al., 2025). In cases where the predictions function like a 'black box', it would be impossible to see the reasons behind the risk categorization (Shantawn et al., 2025). It is because early-warning predictions can materially influence advising and intervention decisions in a high-stakes educational setting. Besides the issue of increasing bias, such lack of transparency is also problematic due to the unaccountability involved (Gunasekara et al., 2025).

This focus is especially important because data-driven systems can reproduce or amplify biases present in training data and lead to unequal outcomes across demographic groups if not carefully designed and monitored (Zhu et al., 2025). Moreover, ethical concerns can arise across the entire data lifecycle, including data collection, analysis, and deployment, highlighting the need for transparency and oversight when predictive models are used to guide educational interventions (Zhu et al., 2025).

Review of the AI Use Case

AI-based early-warning systems are increasingly used in higher education to predict student outcomes, such as pass, fail, or withdrawal, using behavioral and demographic data. These systems analyze learning management system activity, grades, and engagement patterns to identify students at risk and trigger interventions such as tutoring or advising. Predictive learning analytics has therefore become a widely adopted approach to improving retention and student success, with studies showing that academic performance can be predicted early in a semester using interaction data and machine learning models (Akçapınar et al., 2019).

One of the most well-known real-world implementations is the Purdue University Signals early-warning system, which uses demographic information, grades, and behavioral data to classify students as low, medium, or high risk and recommend interventions (Arnold, 2010) . The

system continuously updates risk status throughout the semester based on student activity and performance indicators. Similar predictive systems have also been developed in recent research to identify dropout risk even before students begin courses, using demographic and socioeconomic variables to flag high-risk individuals for early intervention (Carballo-Mendivil et al., 2025). These examples demonstrate that AI-driven student prediction systems are already deployed across universities and continue to expand.

Despite these benefits, predictive early-warning systems raise significant ethical concerns. One major issue is algorithmic bias. Models trained on historical academic data may reproduce existing inequalities related to socioeconomic status, prior education, or demographic factors. For example, dropout prediction systems often rely on variables such as GPA, age, or economic background, which may disproportionately classify disadvantaged students as high risk (Carballo-Mendivil et al., 2025). Such classifications may reinforce stereotypes, influence teacher expectations, and lead to unequal allocation of educational resources. This creates fairness concerns because predictions may affect students' opportunities and academic trajectories.

Real-world controversies in AI-based educational decision-making further illustrate these risks. Universities have increasingly used AI tools for academic evaluation and monitoring, including automated detection and predictive systems. In one reported case, students were falsely accused by AI-based detection tools, creating stress and academic consequences, with evidence that such systems disproportionately flagged non-native English speakers and neurodivergent students (Coldwell, 2024). Although this example concerns AI detection, it demonstrates how algorithmic educational systems can produce unfair outcomes and affect student well-being.

Whereas algorithmic bias is one issue, another major practical and ethical challenge lies in the opacity of many machine learning models. Many early warning systems act as "black boxes," delivering an output in the form of a risk score without disclosing the features that were used in predicting the score (Shantawn et al., 2025). This problem poses significant difficulties for all stakeholders involved. For teachers, obtaining an "at-risk" notification without an explanation would make it challenging to implement a tailored intervention plan (Gunasekara et al., 2025). For students, being subjected to an algorithmically generated profile is not empowering because they cannot appeal or question the results due to the lack of transparency (Shantawn et al., 2025). Thus, incorporating explainable artificial intelligence (XAI) techniques such as SHAP or LIME into early warning systems has become essential to ensure that the black box can be opened and that the outputs can be put into practice ethically (Waseem et al., 2026; Bettahi et al., 2025). Modern findings also indicate that explainability in education prediction models is necessary not only from a technical point of view but also from a social and perceptual perspective. For instance, it was found in academic achievement prediction studies that presentation formality of an explanation may affect the perceived fairness, trustworthiness, and legitimacy of the tool in

question (Lünich et al., 2024). In other words, explainability cannot simply be understood as making the inner workings of the model transparent; it must allow for proper interpretation and questioning of its predictions.

In this case, transparency and explainability become mandatory requirements of the early warning system rather than just options. Lack of interpretability will make it impossible to assess the use of prediction through critical evaluation of the decision support tools or to challenge them from the standpoint of the students. This will not only reduce its effectiveness as a tool to target certain interventions, but will also pose questions about ethical responsibility of those using this kind of prediction tool.

Next Steps

Fairness (Agit Sefkan Balcı) is one of the key AI ethics principles considered in this project. It is especially important in AI-based student dropout prediction because these systems are trained on historical educational data that may already reflect social and structural inequalities. As Baker and Hawn (2022) explain, algorithmic bias in education can emerge when models learn patterns connected to existing disadvantage rather than actual student potential. Anderson, Boodhwani, and Baker (2019) also show that educational prediction systems should not be evaluated only by overall accuracy, since a model may still produce unfair outcomes across different student groups. This concern is further supported by Idowu, Koshiyama, and Treleaven (2024), who investigate algorithmic bias in student progress monitoring and demonstrate that bias can be observed across demographic groups in this context. In this use case, fairness is important because it focuses on reducing unequal outcomes, identifying biased model behavior, and helping ensure that dropout prediction systems support students without reinforcing existing educational inequalities.

The principle of **Transparency & Explainability (Çağrı Altınüzengi)** was chosen to tackle the issues of black box predictions made by artificial intelligence based early warning systems and how this affects critical educational decision making. In predictive analytics for student outcomes, students are categorized as either low risk or at risk based on predictions from algorithms that are not always clear to comprehend. This makes it hard for all parties to grasp the reason behind categorizing a student as “at risk,” which can undermine any intervention efforts and stop students from challenging decisions that could jeopardize their future.

Considering the use case scenario of OULAD, this principle seeks to guarantee that predictions have understandable and accessible explanations for different parties. In the subsequent phase of our project, we intend to develop a system where each prediction made about risks is backed up by interpretable explanations such as feature importance, direction of impact, and concise explanations in plain language. XAI techniques like SHAP and LIME can be considered to

generate instance-wise explanations, but we also reserve the option to explore other methods involving interpretable machine learning models.

More specifically, explainability is not going to be considered only as a technical attribute, but rather as a user-centric demand. The explanations have to be made according to the needs of various groups: instructors can require comprehensive analyses that will help them take actions, and students can profit from more simplistic explanations that empower them. As was found recently, the quality of explanations is determined not by their accuracy alone, but by users' perception of them.

Additionally, this idea will be evaluated based on how it compares to other moral theories. An example is that a description can breach privacy if it involves personal traits, while overgeneralization can lead to misunderstanding. Consequently, this framework will aim to balance transparency, privacy, and ethicality to ensure that the explanation offered is effective and respectful. In summary, integrating the concept of explainability within early warning systems should lead to increased accountability, informed decision making, and human oversight in educational AI.

The **Awareness & Literacy (Yasemin Günindi)** principle was selected to address the limited understanding among students and educators of how AI-based early-warning systems operate and influence educational decisions. In learning analytics systems built on datasets such as OULAD, students may be classified as “at risk” based on behavioral and demographic data without knowing how these predictions are generated or used. This lack of awareness can lead to overreliance on algorithmic outputs, reduced student agency, and limited ability to question automated decisions. AI literacy therefore involves helping stakeholders understand how data are collected, how models generate predictions, and what limitations or biases may exist. This includes both data literacy, understanding how data are collected and analyzed, and algorithm literacy, understanding how models detect patterns and produce decisions. In the OULAD use case, implementing Awareness & Literacy means informing students about data collection practices, explaining how risk predictions are produced, and training educators to interpret predictions critically rather than treating them as objective truth. UNESCO emphasizes that AI literacy in education should empower learners and educators to understand the social and ethical implications of AI systems and avoid overreliance on automated tools. Improving awareness and literacy helps stakeholders interpret predictions more effectively, question potential biases, and maintain meaningful human oversight in educational decision-making.

Privacy (Edoardo Balzano):

Taking into consideration privacy implications in terms of the implementation of the early-warning system is a prerequisite for the protection of students' rights and interests in the context of applying the Open University Learning Analytics Dataset (OULAD). Indeed, since different ethical problems might arise from the beginning of data processing and all its further

stages, from data collection to prediction and implementation, privacy protection measures should be provided. In particular, during the next stage of the project development, privacy implications will be taken into account in order to guarantee that the algorithm will not disclose personal data, especially, in cases where the economic status or age of students are used as a criterion for identifying those at risk. At this point, it is important to highlight that the task of explaining why certain students were classified as at-risk can lead to data breaches because revealing the explanation would mean disclosing personal information of those students. To solve the problem, the privacy framework will focus on the examination of interpretable machine learning models and XAI solutions based on SHAP and LIME approaches. Thus, by explicitly defining what kinds of interaction data in learning management systems will be utilized and how the data will be communicated to students, one will provide explanations in terms of instances without disclosing any private details.

Conclusion

This report examines the possible ethical implications of AI-based systems in education, specifically focusing on early warning systems that predict student outcomes using the OULAD dataset. While AI-based systems can support early intervention and improve retention as discussed, they also introduce risks related to bias, lack of transparency, privacy concerns and limited awareness. The selected principles; fairness, transparency & explainability, and awareness & literacy highlight the importance of ethical and responsible deployment of AI systems. Fairness helps prevent inequalities across different student groups, transparency enables understanding and accountability, and awareness supports the informed interpretation of predictions.

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APPENDIX

Appendix A: Diagram of Implementation

